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Synthesis of Niacin from 3-Cyanopyridine with Recombinant *Escherichia coli* Carrying *afnitA* Nitrilase in a Deep Eutectic Solvent System

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Abstract

Niacin is a compound with a wide range of applications in pharmaceuticals, healthcare, food nutrition, animal breeding, cosmetics, etc. A recombinant *Escherichia coli* carrying the *afnitA* nitrilase gene was created to transform 3-cyanopyridine into niacin in this work. After analyzing the viscosity, surface tension, and Kamlet-Taft (K-T) parameters (π*, α, and β values) of certain deep eutectic solvents (DESs), Betaine:Acetic Acid (Betaine:AA) (1:2, mol/mol) was chosen as the bioreaction medium. Using response surface methodology (RSM), systematic biocatalytic optimization was performed. The optimum medium pH, cell loading, temperature, and DES (Betaine:AA) (1:2, mol/mol) dose were determined to be 7.75, 195 g/L, 44.24 °C, and 18.04 wt%. Under the optimized conditions, whole-cell catalysis facilitated the conversion of 3-cyanopyridine to niacin, achieving a high yield of 98.6% within 40 min. These results demonstrated that recombinant *E. coli* carrying the *afnitA* nitrilase gene may have practical value as a biocatalyst for the production of niacin, with promising prospects for future applications.

Keywords: 3-cyanopyridine; niacin; nitrilase; biosynthesis; deep eutectic solvent

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1. Introduction

Niacin (also known as nicotinic acid or vitamin B3) is a water-soluble vitamin belonging to the B-vitamin family, one which plays key roles in cellular metabolism and energy production [1,2]. In addition to its biological significance, niacin is extensively used in the pharmaceutical and food industries in treatments for pellagra [3,4] and hyperlipidemia [5,6]. It also can be utilized as an antistatic agent, preservative, biocide, chelating agent, and organocatalyst [7,8]. Traditionally, niacin can form through chemocatalysis from quinoline or 3-methylpyridine [9] (Figure 1a) under harsh reaction conditions (high temperature, high pressure), a process which is prone to by-products and therefore one that pollutes the environment. With the increasing demand for niacin, there is a growing need for manufacturing methods that are more efficient and sustainable [8].

Nitrilases (EC 3.5.5.1) are a class of industrially important enzymes that transform nitriles into the corresponding carboxylic acids and ammonia in a single step [10–12]. Biocatalysis is generally defined as the utilization of isolated enzymes or whole cells for the conversion of a series of natural and non-natural chemicals [13,14]. Nitrilase plays significant roles in large-scale biocatalytic processes in the chemical industry [15], owing to

its unique chemical-, regio-, and stereo-selectivity [16]. The enzymatic process of converting 3-cyanopyridine to niacin using nitrilases is an attractive alternative to the chemical route [17] (Figure 1b). The nitrilase catalysis has other advantages, such as good selectivity, few by-products, high product purity, and the lack of a need for coenzymes [18]. Although nitrilases have shown great potential in various biocatalytic applications, their broader industrial implementation is still hindered by challenges such as low catalytic efficiency, poor enzyme stability, and narrow substrate specificity. To overcome these limitations, various strategies have been developed to enhance the catalytic performance of nitrilases, including enzyme engineering, immobilization techniques, optimization of reaction conditions, combinatorial biocatalysis, and chemical modification, as well as metabolic engineering and synthetic biology [19,20]. In this research, we focus on optimizing reaction conditions as a key strategy to improve the catalytic efficiency of nitrilases.

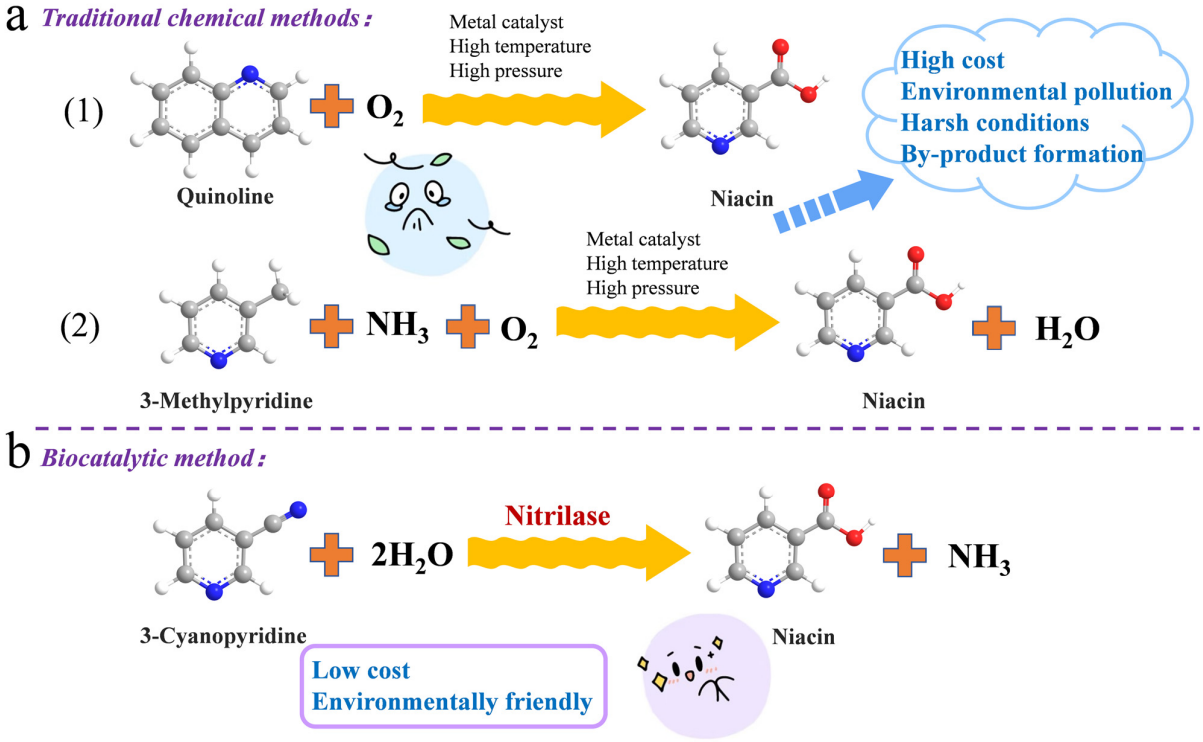


Figure 1. Traditional chemical methods (**a**) and new biological method (**b**).

There have been numerous reports on the efficient catalysis of nitriles into carboxylic acids using nitrilases. Jin et al. [21] utilized immobilization technology to employ nitrilase from *Thalassiosira pseudonana* as a biocatalyst for the efficient catalysis of 2-hydroxy-4-(methylthio) butanenitrile into 2-hydroxy-4-(methylthio) butanoic acid, achieving a high immobilization-based recovery of active enzymes of over 90%. Tang et al. [22] utilized enzyme molecular engineering optimization and employed nitrilase from *Paraburkholderia graminis* as a biocatalyst to convert 2-chloronicotinonitrile into 2-chloronicotinic acid, achieving an optimal conversion rate of 85.7%. Zhang et al. [23] utilized enzyme molecular engineering optimization via enzyme molecular engineering and screened a nitrilase N258D mutant from *Arabis alpina* as a biocatalyst to convert isobutyl succinonitrile into (S)-3-cyano-5-methylhexanoic acid, a key chiral intermediate for pregabalin, achieving a conversion rate of 45%. Dai et al. [10] used recombinant nitrilase RzNIT/W167G from *Rhodococcus zopfii* as a biocatalyst for the whole-cell catalysis of 2-chloronicotinonitrile into 2-chloronicotinic acid, achieving a final 2-CA concentration of 318.5 mM. Nitrilase-

mediated biotransformation has gained much attention with respect to the manufacture of organic acids.

Previous studies have confirmed that a DES can serve as a medium in biocatalytic reaction systems to enhance catalytic performance [24,25]; the medium can be prepared by combining hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs) at a certain stoichiometric ratio [26–28]. The DES was not only easy to prepare but also enhanced the interaction between the substrate and the catalyst during the reaction [29]. As some of the most eco-friendly solvents available, DESs are extensively utilized in organic synthesis, biocatalysis, and electrochemical processes, as well as in the biomedical field [30–33]. A broad variety of enzymes, as well as living microbial cells, have been successfully utilized for biotransformations in DESs [34]. For example, choline chloride:ethylene glycol (ChCl:EG) is relatively easy to prepare, and is currently one of the co-solvents most suitable for improving the efficiency of biotransformations. It has played an important role in the enzymatic catalytic reaction of daidzein [35]. The addition of DESs such as choline chloride:glycerol (ChCl:GL) can enhance the efficiency of lipase-catalyzed esterification reactions [36]. Similarly, the addition of ChCl:GL also enhanced the efficiency of an esterification reaction catalyzed by subtilisin [37]. The addition of DESs such as betaine:glycerol (Betaine:GL) or betaine:sorbitol:water (Betaine:Sor:H_2O) has been proven to be comparatively beneficial for the stabilization of laccase [38]. The addition of betaine:lactic acid (Betaine:LA) can effectively enhance the efficiency of *Pseudomonas putida* S12 in catalyzing the conversion of 5-hydroxymethylfurfural to 2,5-bis(hydroxymethyl)furan [39]. Clearly, DESs can be used as benign reaction media, potentially enhancing the biocatalytic activity of nitrilase biocatalysts.

In this research, recombinant *E. coli* containing the nitrilase gene *afnitA* was constructed and used as a biocatalyst to achieve the bioconversion of 3-cyanopyridine into niacin. The catalytic efficiency of the nitrilase was improved by optimization via single-factor analysis, including consideration of the pH of the PBS buffer, reaction temperature, and cell loading. Additionally, DESs were introduced into the reaction medium, and various binary DESs combinations were screened. The work aimed to develop an efficient biocatalytic process for niacin production in a DES system.

2. Results and Discussion

2.1. Construction of Recombinant E. coli Expressing Nitrilase AfnitA

In this research, a recombinant *E. coli* carrying nitrilase AfnitA was constructed to transform 3-cyanopyridine into niacin efficiently. According to works in the literature [40], employing pRSFDuet as a recombinant vector has the potential to boost the production of nitrilase within *E. coli* cells. The construction process is illustrated in Figure 2a. Since the nitrilase from *Alcaligenes faecalis* MTCC 126 (encoded by the *afnitA* gene) exhibits high catalytic efficiency, enantioselectivity, thermal stability, operational stability, and substrate specificity [41], we chose to insert the *afnitA* gene into vector pRSFDuet-1 to produce pRSFDuet-*afnitA*. This recombinant plasmid was then transformed into *E. coli* BL21, resulting in the complete recombinant strain. The transformed *E. coli* BL21 cells were subsequently transferred to an Luria–Bertani (LB) solid medium containing kanamycin to select for successful transformants. Plasmid extraction, followed by sequencing, confirmed the correct construction of the recombinant plasmid. After confirming the integrity of the construct, the pRSFDuet-*afnitA* was introduced into *E. coli* BL21 competent cells for expression. As shown in Figure 2b, the results of agarose gel electrophoresis indicate that the *afnitA* fragment is 1309 bp in size, further confirming the successful construction of the strain.

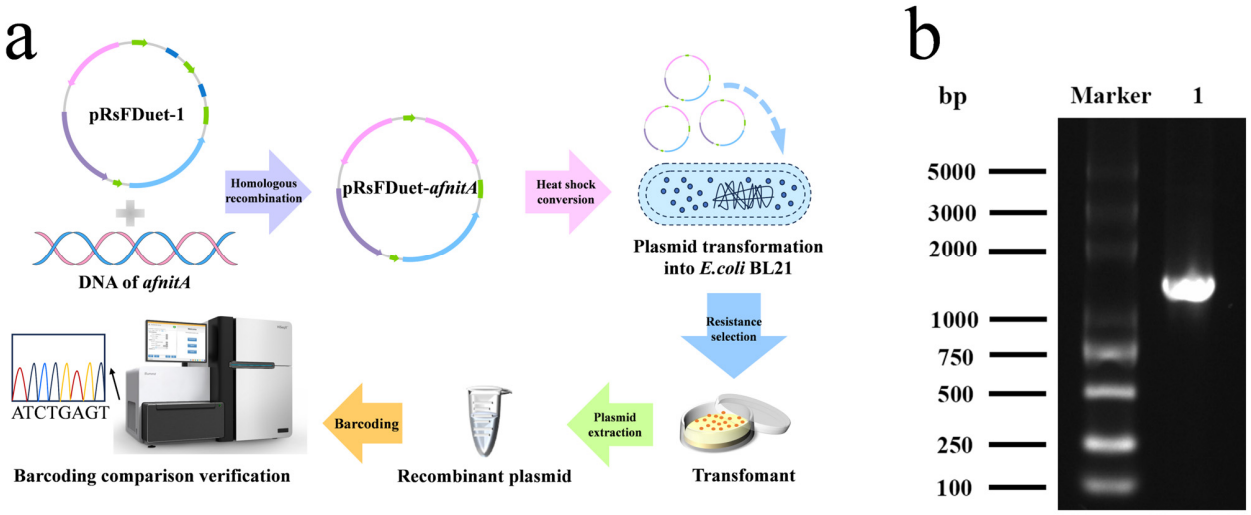


Figure 2. Construction and validation process of recombinant *E. coli* pRSFDuet-*afnitA* (**a**); Agarose gel electrophoresis of *afnitA* gene [left lane: Marker (unit: bp)] (**b**).

2.2. Whole-Cell Biocatalysis by Nitrilase Through Single Factor Analysis

The pH level of the PBS, the temperature, and the cell-mass concentration in reactions are crucial factors influencing catalytic efficiency [42,43]. In practice, precise control of these conditions is decisive for maintaining enzyme activity and ensuring an efficient reaction [44–46]. In this research, the reaction shaking speed of 220 rpm, reaction time of 3 h, and substrate 3-cyanopyridine concentration of 100 mM were set as fixed conditions during the screening of common single factors. Firstly, the effects of medium pH were investigated in this work. Under the conditions of a reaction temperature of 30 °C and cell-mass concentration of 100 g/L, the pH of the PBS (50 mM) was varied between 6.0 and 8.0, and the results are provided in Figure 3a. It was observed that the niacin yield increased as the pH of the PBS was raised from 6.0 to 7.5. Notably, the niacin yield increased sharply when the pH was raised from 6.0 to 6.5, while the growth rate of the niacin yield slowed down as the pH was increased from 6.5 to 7.5. The niacin yield reached its peak at a pH of 7.5. However, when the pH was further increased from 7.5 to 8.0, the niacin yield continuously decreased. This indicated that both overly acidic and overly alkaline conditions might inhibit nitrilase activity. So, the optimal pH for the biocatalytic reaction was determined to be 7.5. Furthermore, the effect of bioreaction temperature was evaluated in light of the biocatalytic reaction efficiency, under the conditions of a PBS with a pH of 7.5 and a cell-mass concentration of 100 g/L. The reaction temperature was then set to a range of 15 °C to 65 °C, as illustrated in Figure 3b. Upon raising the temperature from 15 °C to 37 °C, the niacin yield was enhanced, with nitrilase activity peaking at 37 °C. As the temperature exceeded 37 °C, the niacin yield began to decrease. Simultaneously, the niacin yields at 15 °C and 65 °C were both very low. This phenomenon indicates that both excessively low and high temperatures can inhibit nitrilase activity. In conclusion, the optimal temperature for the biocatalytic reaction was determined to be 37 °C. Moreover, the effect of cell-mass concentration was examined with respect to the niacin production, under the fixed conditions of a PBS with a pH of 7.5 and a reaction temperature of 37 °C. The cell-mass concentrations across the different experiments ranged from 20 to 200 g/L, as illustrated in Figure 3c. As the cell-mass concentration was increased from 20 to 150 g/L, the niacin yields improved. Specifically, the optimal cell-mass concentration was 150 g/L. When the cell-mass concentration was further increased to 200 g/L, the niacin yield slightly decreased, suggesting that high levels of bacterial concentration may inhibit enzyme activity. Thus, the suitable cell-mass concentration was 150 g/L. As a concise summary, it

was concluded that the optimal conditions given by single-factor optimization at a substrate concentration of 100 mM, a shaking rate of 220 rpm, and a reaction time of 3 h were as follows: a PBS pH of 7.5, a reaction temperature of 37 °C, and a cell-mass concentration of 150 g/L.

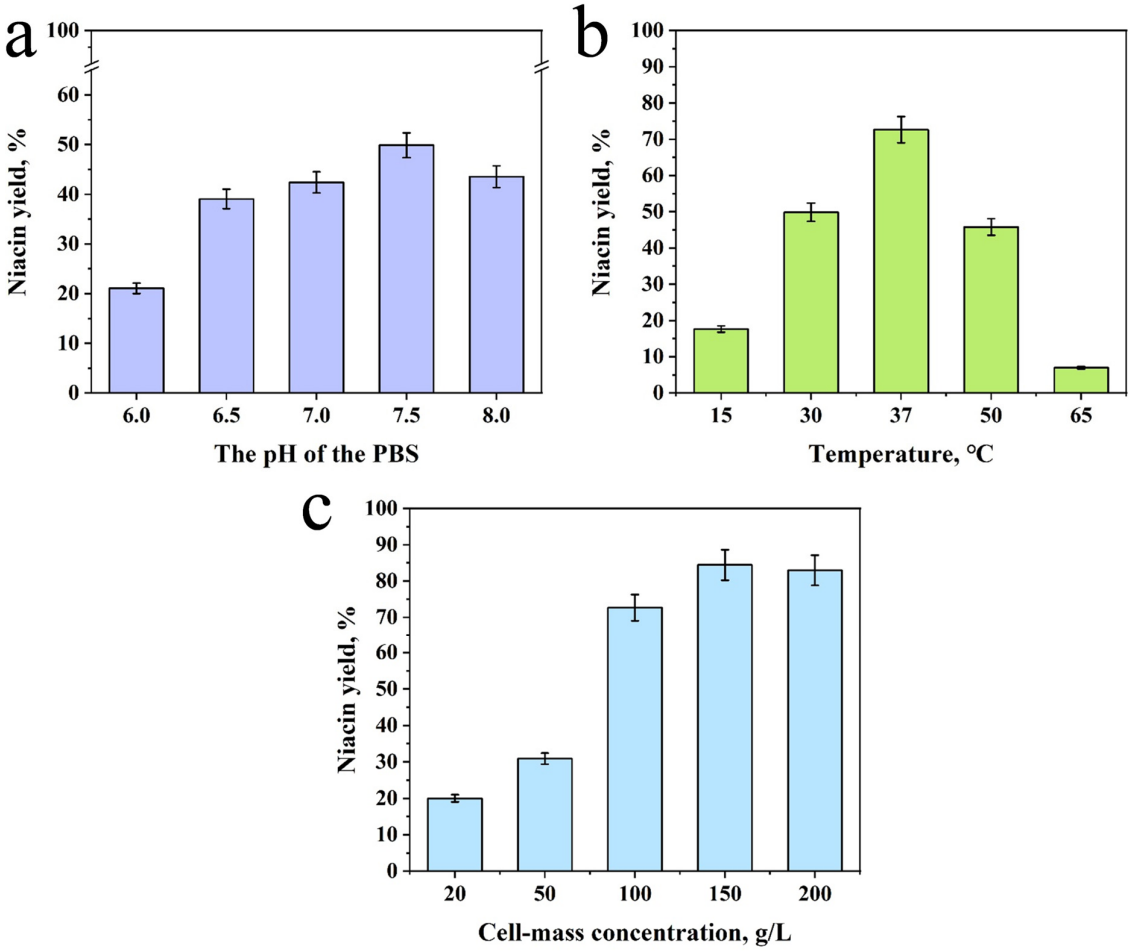


Figure 3. Effects of the pH of the PBS (6.0–8.0) (**a**), temperature (15–65 °C) (**b**), and cell-mass concentration (20–200 g/L) (**c**) in the transformation of 3-cyanopyridine to niacin.

Ecofriendly solvents are used in many chemical and biological reactions because of their efficient and environmentally friendly characteristics [47], and the addition of appropriate solvents to improve the catalytic efficiency of nitrilase is an innovative case in point. It has been demonstrated that the addition of a DES may favor whole-cell biocatalysis and increase the permeability of cell membranes [48]. To screen for the most effective DES in maximizing the catalytic efficiency of nitrilase, four acids (lactic acid, propionic acid, acetic acid, and formic acid) and two alcohols (glycerol and ethylene glycol) were selected. Additionally, two bases, choline chloride and betaine, were chosen. A total of twelve binary DESs were obtained by mixing the acids with the bases, or the alcohols with the bases, in specific ratios. Choline chloride was mixed with all the acids and the alcohols, in a ratio of 1:2, and betaine was mixed with all the acids and the alcohols, also in a ratio of 1:2, and the process produced clear, transparent liquids.

To further investigate the characteristics of the DESs, the parameters of different DESs were measured, as presented in Table 1. Among the twelve types of DESs, Betaine:GL exhibited the highest viscosity. As displayed in Figure 4a, compared with the control, the catalytic efficiency of Betaine:GL was relatively lower. This could be attributed to the high viscosity, which may increase the resistance involved with the biocatalyst's contact with

the substrate and thereby affect the biocatalytic efficiency. Surface tension refers to the minimum energy required to form or expand a unit interface; this can be used to express surface free energy [49]. Among the tested DESs, Betaine:LA exhibited the highest surface tension, indicating that it possessed the highest surface free energy. By analyzing the values of π*, α, and β in the K-T parameters of the DES, the relationship between the K-T parameter values and the structure and properties of DESs can be established [50]. The K-T parameters are a set of parameters used to describe solvent polarity and hydrogen-bonding properties, and which help to understand and predict the behaviors of solutes in different solvents, such as solubility, reaction rates, and spectroscopic characteristics. Among these parameters, π* reflects the polarity of the DES, α denotes the solvent's ability to donate protons and form hydrogen bonds with solutes, and β indicates the hydrogen-bond basicity of the solvent. A higher β value suggests that the DES can more readily accept hydrogen bonds [51]. Notably, ChCl:FA exhibits the highest α value, indicating that it demonstrates the strongest ability to donate protons and form hydrogen bonds. Betaine:GL has the highest β value, indicating that it is the tested DES most likely to accept hydrogen bonds.

Table 1. Different DES parameters and K-T parameters.

Constituents	HBA/HBD Molar Ratio	Viscosity (mPa·S)	Surface Tension (mN/m)	K-T Parameters		
				π*	α	β
ChCl:LA	1:2	137.11	45.60	0.80	0.69	0.48
ChCl:PA	1:2	97	31.09	0.77	0.55	0.71
ChCl:AA	1:2	59	29.48	0.53	0.64	0.43
ChCl:FA	1:2	88.38	44.88	0.43	1.7	0.76
ChCl:GL	1:2	284.89	44.62	1.08	0.65	0.88
ChCl:EG	1:2	89	43.68	1.03	0.68	0.87
Betaine:LA	1:2	376	58.06	1.37	0.59	0.67
Betaine:PA	1:2	232.6	33.76	1.24	0.9	0.64
Betaine:AA	1:2	488	41.03	1.34	0.84	0.53
Betaine:FA	1:2	274.84	57.70	1.26	0.73	0.78
Betaine:GL	1:2	2623.22	49.82	1.06	0.62	0.97
Betaine:EG	1:2	104	51.70	1.03	0.64	0.51

Different experiments were carried out involving the addition of twelve types of DES, at concentrations of 25 wt%, under the following conditions: substrate concentration of 100 mM and with the use of ethanol as a co-solvent, PBS with a pH of 7.5, cell-mass concentration of 150 g/L, shaking rate of 220 rpm, reaction temperature of 37 °C, and reaction time of 1 h. The reaction was carried out at a concentration of 25 wt% for each of the 12 types of DES. As showcased in Figure 4a, the addition of 25 wt% of a newly synthesized DES was shown to improve the catalytic efficiency of nitrilase compared to the results obtained without the addition of a DES. Adding DES (Betaine:AA) could improve niacin yield significantly, while ChCl:EG, Betaine:GL, and Betaine:EG each showed an inhibitory effect on the catalytic efficiency of nitrilase, indicating that the addition of the vast majority of species of DES might improve the catalytic efficiency of nitrilase. Consequently, the addition of DES (Betaine:AA) was selected as the process best suited to improve the niacin yield.

Focusing specifically on the effect of DES (Betaine:AA) on the catalytic activity of nitrilase, seven concentration gradients were tested: 0 wt%, 2 wt%, 5 wt%, 10 wt%, 15 wt%, 20 wt%, and 25 wt%. As the concentration of DES was elevated from 0 wt% to 15 wt%, the increase in niacin yield continuously accelerated, and the catalytic performance of nitrilase reached its highest when the DES concentration was 15 wt% (Figure 4b). As the

DES concentration increased from 15 wt% to 25 wt%, the niacin yield demonstrated no significant change. Therefore, the medium containing 15 wt% of DES (Betaine:AA) was used as the optimum bioreaction system.

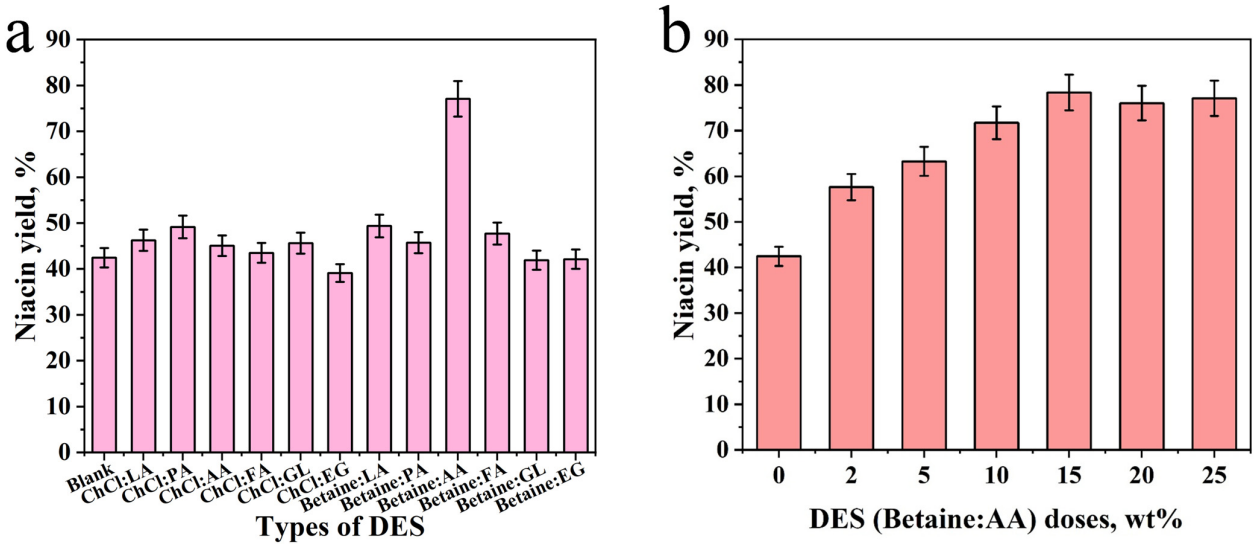


Figure 4. Effects of doses of the different types of DES (25 wt%) (**a**) and DES (Betaine:AA) (wt%) (**b**) on the transformation of 3-cyanopyridine to niacin.

2.3. Optimization of the Nitrilase-Mediated Reaction via Response Surface Analysis

To maximize the enzyme yield of nitrilase from recombinant *E. coli* pRSFDuet-*afnitA* as well as to establish an efficient nitrilase catalytic system and optimize its reaction parameters, the Box–Behnken method in RSM was used in this research to optimize the design of key parameters. Based on the previously described common single-factor optimization and considering the addition of a DES to improve the catalytic activity of nitrilase, the following experimental ranges were designed and tested: (A) pH of PBS (7.0–8.0), (B) Cell-mass concentration (100–200 g/L), (C) Reaction temperature (30–50 °C), and (D) DES (Betaine:AA) doses (10–20 wt%). A total of 29 sets of nitrilase-catalyzed conversions of 3-cyanopyridine to niacin were performed, and the results are stated in terms of the efficiency of the niacin conversion, as obtained after 30 min of reaction, and the co-solvent for solubilizing the substrate 3-cyanopyridine was ethanol.

The ANOVA in Table 2 shows a p-value < 0.01 for the model term, indicating a significant relationship between transformation efficiency and the regression equations of the influencing factors, along with a high degree of confidence. Conversely, a p-value > 0.05 for the lack-of-fit term indicates the absence of a significant difference, suggesting that the proportion of the non-normal error in the obtained equation relative to the actual fit is small and that the test results are well-fitted to the regression model. The primary analysis revealed statistically significant impacts (p-value < 0.01) for all four factors (A: The pH of the PBS, B: Cell-mass concentration, C: Reaction temperature, and D: DES doses) relative to transformation efficiency. The hierarchy of influence can be ranked as B > D > A > C, indicating that cell-mass concentration (B) exerted the dominant effect. In the secondary analysis, quadratic terms A^2 and C^2 also demonstrated significant contributions to transformation efficiency (p-value < 0.01). The efficiency and adequacy of the model were verified using ANOVA. The F-value calculated for the model was 8.47, confirming its significance. The coefficient of determination (R^2) was 0.8944, indicating that the mathematical model explained 89.44% of the experimental data. Additionally, the adjusted R^2 (R^2adj) was 0.7889, suggesting that 78.89% of the variability in the data was accounted for by the model.

Table 2. The analysis of variance (ANOVA) for the Box–Behnken model.

Source	Sum of Squares	df	Mean Square	F-Value	*p*-Value
Model	6950.75	14	496.48	8.47	0.0001
A (The pH of the PBS)	768.80	1	768.80	13.12	0.0028
B (Cell-mass concentration)	2846.23	1	2846.23	48.57	<0.0001
C (Temperature)	551.08	1	551.08	9.40	0.0084
D (DES doses)	1004.30	1	1004.30	17.14	0.0010
AB	25.30	1	25.30	0.4317	0.5218
AC	119.25	1	119.25	2.03	0.1756
AD	0.7656	1	0.7656	0.0131	0.9106
BC	0.0121	1	0.0121	0.0002	0.9887
BD	315.24	1	315.24	5.38	0.0360
CD	18.23	1	18.23	0.3111	0.5858
A^2	898.04	1	898.04	15.32	0.0016
B^2	1.18	1	1.18	0.0201	0.8892
C^2	561.04	1	561.04	9.57	0.0079
D^2	23.36	1	23.36	0.3986	0.5380
Residual	820.41	14	58.60		
Lack of fit	412.27	10	41.23	0.4040	0.8882
Pure error	408.14	4	102.04		
Cor total	7771.16	28			

The effect of each parameter on the relative activity of nitrilase was visualized using three-dimensional response surface plots. The relative activity of nitrilase, as reflected by conversion efficiency, was analyzed through a four-factor analysis involving the pH of the PBS, cell-mass concentration (g/L), reaction temperature (°C), and DES (Betaine:AA) dose (wt%). These response surface plots illustrated not only the individual effects of these variables but also their interactions [52]. As displayed in Figure 5a,b, in the results of the two-factor analysis of catalytic efficiency that considered the pH of the PBS/DES (Betaine:AA) dose and the temperature, it can be observed that when the pH of the PBS or DES (Betaine:AA) dose was kept constant, the conversion efficiency tended to increase with the increase in temperature; the efficiency initially rose more rapidly and then tended to level off at a later stage, suggesting that the lower temperature inhibited the nitrilase activity, while the activity of nitrilase was more stable when the temperature reached 40–50 °C. Under a certain temperature, with the increase in the pH of the PBS, the conversion efficiency showed a tendency to increase and then decrease, and the initial increase was rapid; additionally, the increase was slow when the pH was close to 7.5, which indicated that a pH in the range of 7.5–8.0 did not have much of an effect on the activity of the nitrilase. These findings collectively validated the robustness of the experimental methodology employed. When the temperature was kept constant, the conversion efficiency continued to increase as the addition of DES (Betaine:AA) was increased within the range of 10–20%, indicating that an increase in DES dosage within a certain range is beneficial to the activity of nitrilase.

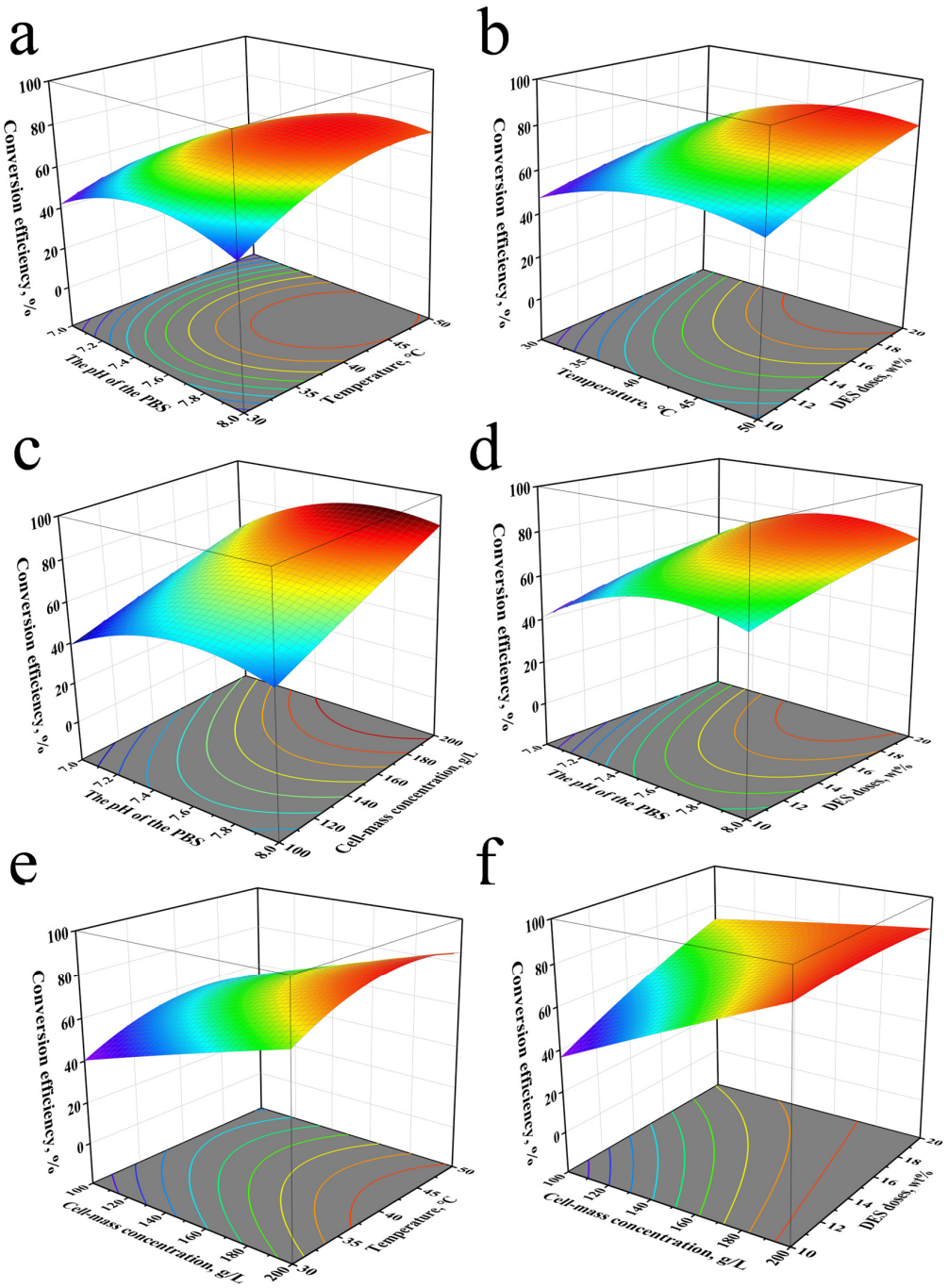


Figure 5. Response surface modeling of the influence of the pH of the PBS (7.0–8.0) and temperature (30–50 °C) (**a**); the influence of temperature (30–50 °C) and DES doses (10–20 wt%) (**b**); the influence of the pH of the PBS (7.0–8.0) and cell-mass concentration (100–200 g/L) (**c**); the influence of the pH of the PBS (7.0–8.0) and DES doses (10–20 wt%) (**d**); the influence of cell-mass concentration (100–200 g/L) and temperature (30–50 °C) (**e**); and the influence of cell-mass concentration (100–200 g/L) and DES doses (10–20 wt%) (**f**).

Through the two-factor analysis considering cell-mass concentration/DES (Betaine:AA) doses and the pH of the PBS (Figure 5c,d), it could be observed that the conversion efficiency of the 3-cyanopyridine changed into niacin was gradually elevated when the pH of the PBS was increased from 7.0 to 7.7 at a certain cell-mass concentration or DES dose, which can be attributed to the fact that key residues of the active site require a higher pH to deprotonate and form the correct ionic state. Thus, such state might facilitate the catalytic reaction or an alteration in the protonation state of the substrate that is more favorable

for binding. The alteration in protonation state might affect protein folding, active site function, and enzyme stability. In strongly alkaline reaction environments, changes in charge distribution around enzyme molecules might induce alterations in the enzyme structure's ionization state relative to amino acid residues [53,54]. This change in ionization state would disrupt the three-dimensional conformation of the enzyme, and thus severely affect its biocatalytic capacity [55].

As presented in Figure 5e,f, when the temperature and the dose of DES were kept constant, the overall trend of the conversion efficiency was enhanced as the cell-mass concentration was raised within the range of 100 g/L to 200 g/L. This phenomenon indicated that an increase in cell loading within the aforementioned range might be beneficial to the activity of nitrilase. This occurred due to the fact that the substrate 3-cyanopyridine as a small molecule could freely diffuse through the cell membrane with good permeability and low mass transfer resistance within the whole-cell catalytic system. Meanwhile, it was demonstrated under the combined influence of various factors that the effect on the nitrilase-catalyzed system was most significant when the cell-mass concentration was 195 g/L. According to the calculation results for the cubic polynomial model, Equation (1), the optimal pH of the PBS was 7.75, the optimal cell-mass concentration was 195 g/L, the optimal reaction temperature was 44.24 °C, and the optimal dose of DES (Betaine:AA) was 18.04 wt%.

$$\begin{aligned} &\textit{Conversion efficiency}(\%) \\ &\quad = 2598.31783 + 665.84833 \times A + 0.132937 \times B - 0.729200 \times C \\ &\qquad +9.03787 \times D + 0.100600 \times A \times B + 1.09200 \times A \times C \\ &\qquad -0.175000 \times A \times D + 0.000110 \times B \times C - 0.035510 \times B \times D \\ &\qquad +0.042700 \times C \times D - 47.06567 \times A^2 - 0.000171 \times B^2 \\ &\qquad -0.093002 \times C^2 - 0.075907 \times D^2 \end{aligned} \quad (1)$$

To verify the accuracy of the model's predictions, the activity of nitrilase was tested under the optimal conditions contemplated by the model described in Equation (1). The model predicted a conversion efficiency of about 91.7%, which was very close to the actual conversion efficiency of 90.9%. This similarity confirmed the reliability of the conversion efficiency obtained under the optimal conditions contemplated by the model.

To evaluate the biocatalytic activity of recombinant *E. coli* pRSFDuet-*afnitA* cells under optimal conditions, an experiment was conducted with the following parameters: PBS buffer pH 7.75, cell-mass concentration 195 g/L, temperature 44.24 °C, DES (Betaine:AA) 18.04 wt%, and ethanol (as co-solvent) 10 wt%. As shown in Figure 6, the yield of the niacin produced increased positively with the consumption of the substrate (100 mM), reaching a maximum yield of 98.6% after 40 min of reaction.

Numerous studies have been reported on the nitrilase-catalyzed conversion of 3-cyanopyridine to niacin. In recent years, researchers have explored various strategies to enhance the catalytic efficiency of nitrilase in this transformation. However, the enzyme's activity is significantly inhibited at high concentrations of 3-cyanopyridine. Although several approaches have been developed to improve substrate tolerance, the inherent inhibition of wild-type nitrilases under high substrate loads, along with product-induced deactivation, remains a major bottleneck for industrial scale-up. In early studies, Jin et al. [56] (2013) demonstrated that resting cells of *Fusarium proliferatum* ZJB-09150 could completely convert 3-cyanopyridine to niacin within 15 min at pH 9.0 and 30 °C, but only when the substrate concentration was $\leq$60 mM, achieving a yield of 98.9%; higher concentrations led to severe inhibition. To overcome this limitation, subsequent efforts shifted toward fed-batch strategies. Badoei-Dalfard et al. [57] (2016) achieved a total substrate loading of 945 mM (approximately 100 g/L) by supplementing 70 mM of substrate every 40 min using

highly induced resting cells of *Stenotrophomonas maltophilia* AC21 (10 U mL^{-1}), resulting in a 90% yield. More recently, Monika et al. [17] (2022) further optimized the induction and feeding protocol using *Gordonia terrae* MTCC8139 resting cells (10 U/mL) with staged induction (0/16/24 h, 0.5% isobutyronitrile) and fed-batch operation, adding 75 mM substrate every 15 min for 22 cycles. This enabled a total substrate concentration of 1.65 M (approximately 202 g/L) with 100% conversion (202 g niacin) and a remarkable space–time yield of 15.3 g h^{-1} g dcw^{-1}—71% higher than the original process—and shortened the reaction time to 5.5 h. In this experiment, through systematic optimization of reaction conditions, we achieved a highly efficient conversion of 3-cyanopyridine to niacin at a single-batch concentration of 100 mM, with a yield of 98.6%. This represents a significant increase in substrate tolerance and surpasses previous concentration limitations under batch conditions, offering a more feasible and scalable pathway for industrial application.

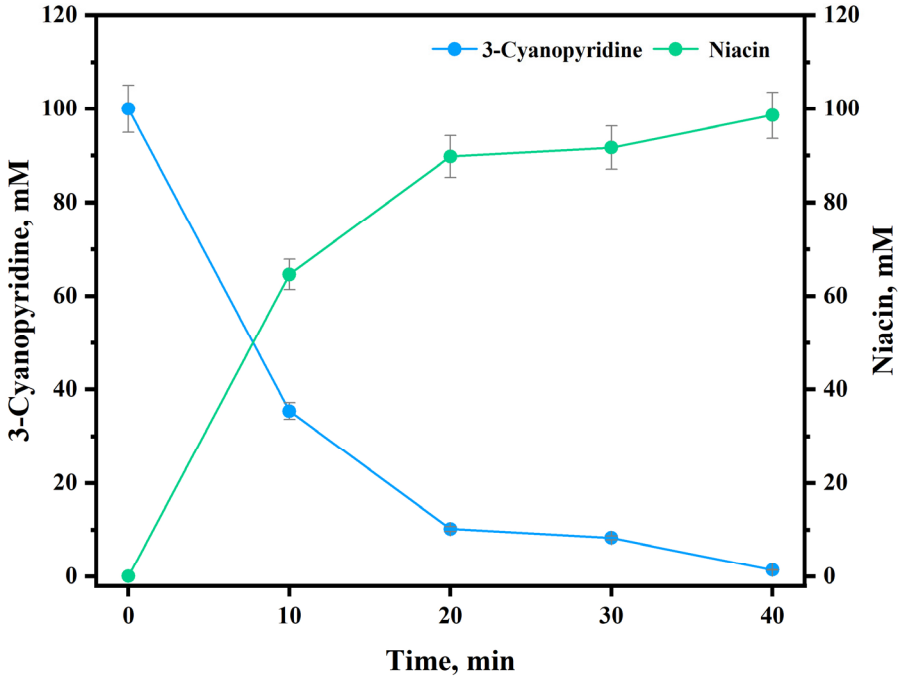


Figure 6. Time course for niacin synthesis from 3-cyanopyridine by AfnitA whole-cell catalysis [reaction conditions: 100 mM 3-cyanopyridine in PBS buffer (pH 7.75) with 195 g/L cell-mass concentration, 18.04 wt% DES (Betaine:AA), 10 wt% ethanol as co-solvent, and reaction temperature at 44.24 °C].

This research aimed to enhance the catalytic activity of nitrilase by optimizing various factors affecting its catalysis, including pH, temperature, co-solvent, and cell loading, as briefly illustrated in Figure 7. The pH value of the bioreaction system is crucial in affecting the activity of nitrilase, and changes in pH can impact the protonation state of catalytic residues, enzyme structural stability, and substrate binding efficiency [58]. Moreover, the temperature of the reaction system significantly influences the activity of nitrilase. Excessively high and low temperatures can affect enzyme activity. High temperatures may cause protein denaturation, disrupting the three-dimensional structure of the enzyme, especially the fine conformation of the active site, leading to loss of activity or even irreversible deactivation. Low temperatures reduce the frequency of molecular collisions and decrease the reaction rate, and thus inhibit enzyme activity [59,60]. Solvent systems are widely used in enzymatic reactions to optimize enzyme activity, stability, and substrate selectivity by adjusting the solvent environment [61–63]. Thus, the introduction of a DES can improve mass transfer efficiency in solvents. Compared to some traditional

ionic liquids, a DES often has lower viscosity, which facilitates increased relative movement speed between molecules, thereby enhancing mass transfer rates. The whole-cell loading of nitrilase biocatalyst also critically impacts its catalytic activity. When the cell dose is low, the insufficient quantity of biocatalyst results in fewer available active sites for catalysis, slowing down the overall reaction rate. By optimizing these factors, highly efficient conversion of 3-cyanopyridine to niacin was implemented through the whole-cell catalysis. In the future, more strategies will be implemented, such as immobilized enzyme technology and directed molecular modification techniques, to address the issue of low nitrilase activity and achieve cost-effective conversion of 3-cyanopyridine to niacin. Although this work primarily focused on improving the synthesis efficiency of niacin, its practical application still requires consideration of product separation and purification. In future work, downstream purification methods such as crystallization (taking advantage of the low solubility of niacin under acidic conditions), solvent extraction, or ion-exchange chromatography could be explored to obtain a high-purity product and facilitate the industrial application of this process.

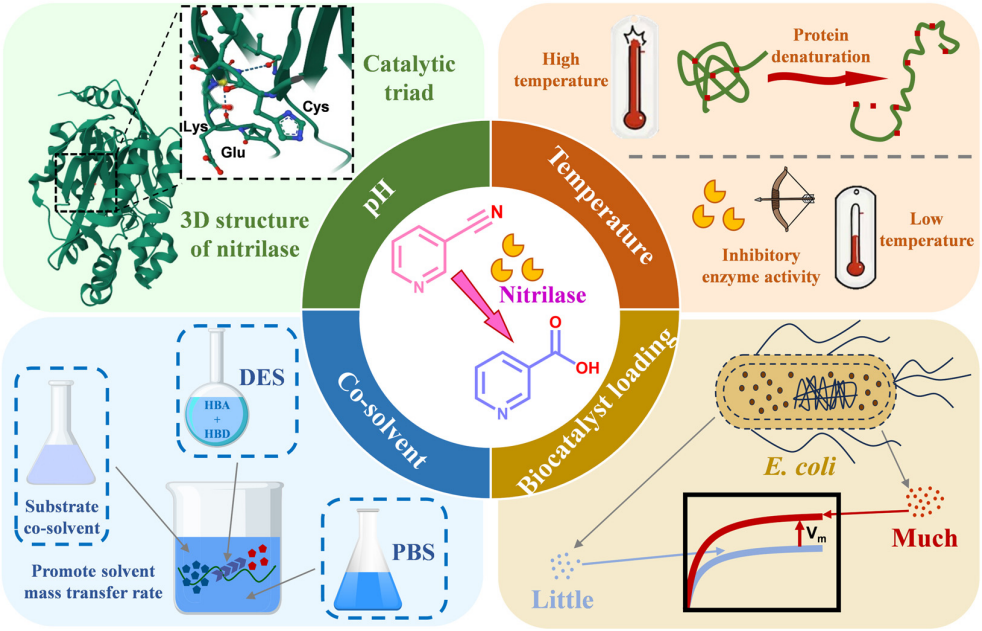


Figure 7. Optimizing various factors affecting nitrilase activity, including pH, temperature, co-solvent, and biocatalyst loading.

3. Materials and Methods

3.1. Materials and Reagents

3-Cyanopyridine and niacin were obtained from Shanghai Adamas Reagent Co., Ltd. (Shanghai, China), while the rest of the nutrients required for the growth of the organisms, including typtone, sodium chloride, and other reagents, were purchased from Shanghai McLean Biochemical Co., Ltd. (Shanghai, China). Choline chloride, betaine, lactic acid, acetic acid, and other reagents were purchased from Shanghai Leadfun Chemical Reagent Co., Ltd. (Shanghai, China).

3.2. Preparation of the DES

In addition to traditional HBAs, other HBAs that offer extra functionalities may be more suitable options [64]. In this research, DESs were prepared using two bases as HBAs, combined with specific HBDs such as acids or alcohols in certain ratios, and formed under appropriate conditions. HBAs (choline chloride and betaine) and HBDs (lactic acid,

propionic acid, acetic acid, formic acid, glycerol, and ethylene glycol) were mixed in a 1:2 molar ratio, resulting in clear, transparent liquids.

3.3. Construction and Culture of Nitrilase AfnitA Engineering Strain

The nitrilase gene *afnitA* [41], derived from *Alcaligenes faecalis* MTCC 126, was synthesized. The expression vector pRSFDuet-1 and the expression strain *E. coli* BL21 were maintained independently. The synthesized *afnitA* gene was ligated into the *EcoRI* and *NcoI* sites of pRSFDuet-1 via homologous recombination. The recombinant vector was then introduced into *E. coli* BL21 competent cells by heat-shock transformation [65]. Transformants were selected by spreading them on LB solid medium plates containing kanamycin (final concentration of 50.0 mg/L). The selected transformants were subsequently verified by sequencing.

The microorganism was cultured and harvested as previously reported [66]. In the reaction vessel, PBS buffer, substrate, and the bacterial suspension were sequentially added to form the reaction system. The reaction conditions were set at a speed of 220 rpm, and the temperature and time were adjusted according to the specific requirements for conducting the biocatalytic reaction.

3.4. Single-Factor Analysis

A total of five single factors were investigated in this experiment on the catalytic activity of nitrilase. Five gradients were set up, at 6.0, 6.5, 7.0, 7.5, and 8.0, to evaluate the influence of the pH of the PBS on catalytic activity. To assess the effect of the cell-mass concentration on the catalytic reaction, five gradients were set up: 20 g/L, 50 g/L, 100 g/L, 150 g/L, and 200 g/L. To assess the effect of temperature on the catalytic reaction, Five gradients were set up: 15 °C, 30 °C, 37 °C, 50 °C, and 65 °C, respectively. To evaluate the effect of the DES on the catalytic reaction, two bases (choline chloride and betaine) and four acids (lactic acid, propionic acid, acetic acid, and formic acid), along with two alcohols (glycerol and ethylene glycol), were mixed in various combinations. The DES with the best catalytic performance was identified as the mixture of betaine and acetic acid in a molar ratio of 1:2. To assess the effect of DES (Betaine:AA) (wt%) on the catalytic reaction, seven gradients were set up, at 0 wt%, 2 wt%, 5 wt%, 10 wt%, 15 wt%, 20 wt%, and 25 wt%, respectively.

3.5. Box–Behnken Design Analysis

RSM can be employed to screen the single factors affecting the biocatalytic reaction. Specifically, the Box–Behnken Design (BBD), which is extensively utilized in response surface studies, can be utilized to achieve the optimum operation conditions [67]. BBD facilitates the determination of key variables within a specified domain through a limited number of experimental runs [68]. Additionally, it supports the forecasting of optimal variable settings to attain the desired outcome [69]. Design-Expert software version 13 (Stat-Ease, Inc., Minneapolis, MN, USA), a professional tool frequently referenced in the literature in the context of optimizing experimental designs, was applied to investigate the factors influencing the three levels of the catalytic reaction: the pH of the PBS, cell-mass concentration, bioreaction temperature, and DES dose. A four-factor relationship was established between the independent variables and their corresponding levels as follows: A, The pH of the PBS—7, 7.5, and 8; B, cell-mass concentration (g/L)—100, 150, and 200; C, Reaction temperature (°C)—30, 40, and 50; and D, Betaine:AA (wt%)—10, 15, and 20. Twenty-nine experiments were conducted randomly to analyze the reaction pattern and model the relative activity of nitrilase. This approach not only minimized experimental errors but also provided robust data for RSM analysis, facilitating the understanding of the effects of each factor on the biocatalytic activity.

3.6. Analytical Methods

The DESs were evaluated by measuring various parameters, including viscosity, surface tension, and K-T parameters—π*, α, and β. To eliminate the influence of the use of different dyes on the K-T parameter values and ensure the comparability of the results, all DESs were uniformly measured using *N*,*N*-diethyl-4-nitrosoaniline (NEt_2) in order to determine π*, Nile red for α, and *p*-nitroaniline (NH_2) for β [70]. The viscosity was measured using an NDJ-5S digital viscometer (Shanghai Yarong Biochemical Instrument Factory, China), and the surface tension was analyzed using a BZY-2 automatic surface tensiometer (Shanghai Balance Instrument Factory, Shanghai, China).

The concentrations of the substrate 3-cyanopyridine and the product niacin were analyzed by high-performance liquid chromatography (HPLC), using an Acclaim 120 C18 analytical column (5 μm × 250 mm × 4.6 mm). Methanol: 0.1 wt% formic acid (30:70, *v*/*v*) was used as the mobile phase at a flow rate of 1 mL/min at 30 °C. Detection of the substrate 3-cyanopyridine and the product niacin was carried out at 264 nm. The concentrations of niacin was calculated using the following equation:

$$Niacin\ yield = \frac{Niacin\ produced\ (mM)}{Initial\ 3-Cyanopyridine\ (mM)} \times 100\% \quad (2)$$

4. Conclusions

This research developed a novel whole-cell nitrilase catalytic process using DES (Betaine:AA) as the reaction medium for the efficient conversion of 3-cyanopyridine to niacin, without the need for additional cofactors or expensive catalysts. Using recombinant *E. coli* carrying the nitrilase AfnitA as a biocatalyst, a hybrid optimization strategy combining one-way analysis and RSM was also employed to optimize the conditions; the final conditions obtained were a pH of 7.75 for the PBS, cell-mass concentration of 195 g/L, temperature of 44.24 °C and dose of DES (Betaine:AA) of 18.04 wt%. 3-Cyanopyridine could be efficiently hydrolyzed to niacin in 40 min with a yield of 98.6%. This work provides an effective strategy for the biocatalytic production of niacin from 3-cyanopyridine.

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